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Perceiving AI as labor-replacing reduces democratic legitimacy and political engagement

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Abstract

AI is expected to reshape society and labor markets, yet experts remain divided on whether AI will primarily displace human labor or generate new employment opportunities. Despite the importance of this debate, little is known about how the public perceives AI's labor market impact—and how these perceptions affect democratic attitudes and behaviors. Large-scale survey data (N = 37,079; 38 European countries) indicate that the public tends to view AI as labor-replacing rather than labor-creating. Controlling for technology-related, political, and sociodemographic factors, these data further show that perceiving AI as labor-replacing (vs. labor-creating) is associated with lower satisfaction with democracy and political engagement with technology. Two preregistered, nationally representative experiments (N = 1,202, United Kingdom; replication study N = 1,200, United States) provide causal evidence for this relationship. Participants exposed to a labor-replacing (vs. labor-creating) AI frame report greater erosion of trust in democracy and lower willingness to politically engage with future AI developments. Together, our findings suggest that perceptions about AI's labor market consequences—regardless of actual outcomes—may decrease democratic legitimacy and public engagement in shaping the future of AI.

Keywords: AI, democratic legitimacy, political engagement

AI is transforming economies and labor markets, but experts remain divided over whether it will ultimately replace or create jobs (1, 2). Although this debate has received considerable attention across the social sciences and media, there is a lack of systematic research on how the public perceives AI's labor market impact—specifically, whether people view AI as labor-replacing or labor-creating and how these perceptions affect broader political attitudes and outcomes (3). These consequences may be significant even though AI's current labor market impact remains limited (1). When AI is perceived as labor-creating, people may be more likely to support its adoption, potentially driving productivity gains across the economy (4). However, when AI is perceived as a threat to jobs, it could lead to broader negative consequences. Prior research shows that perceived threats to employment—such as those from globalization or automation—can undermine democratic satisfaction and fuel populist support (5, 6). The legitimacy of democratic institutions, the belief that democratic institutions are appropriate and responsive to societal needs (7), often rests on their perceived ability to provide economic security, particularly through stable employment (8). If people view AI as a threat to economic security—and believe that democratic institutions are unresponsive to this threat—they may lose trust in those institutions and disengage politically. Yet, empirical, cross-national evidence on these relationships remains scarce (3).

We examine i) whether the public views AI as labor-replacing rather than labor-creating, and ii) how these perceptions affect two core democratic outcomes: democratic legitimacy (e.g., satisfaction with democracy; Ref. 9) and political engagement (e.g., involvement in shaping AI policy; Ref. 10). The latter question is especially timely, as the future trajectory of AI—and its impact on labor—depends not only on technological progress but also on public engagement in AI governance and regulation (11).

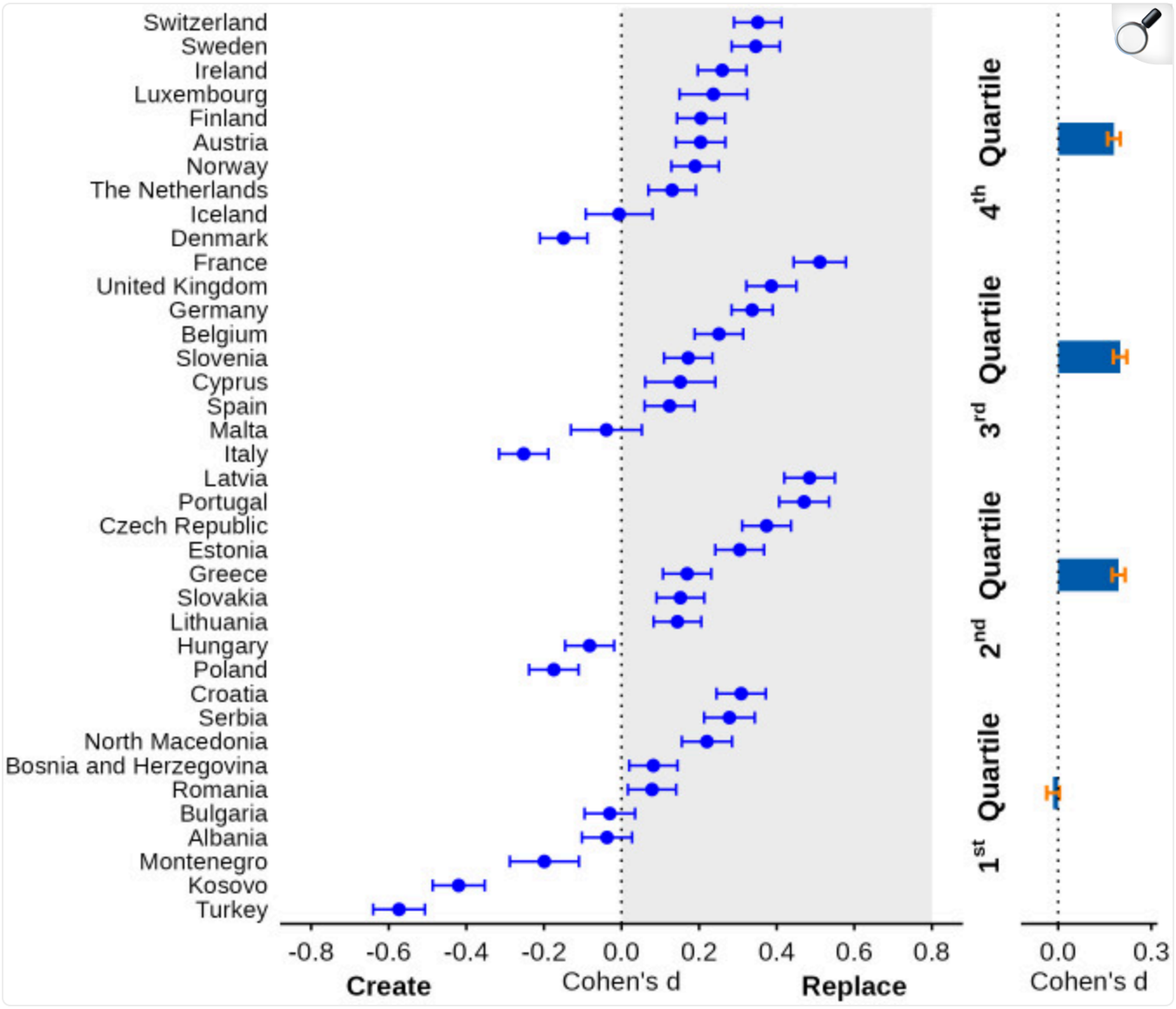
We tested our ideas with two complementary studies. First, we analyzed survey data from 38 European countries (N = 37,079) to assess public perceptions of AI's labor market impact and their relationship with democratic legitimacy and political engagement. Second, we conducted preregistered experiments with nationally representative UK and US samples (total N = 2,402) to test whether these perceptions causally affect the same two democratic outcomes. Together, our findings suggest that perceptions of AI's labor market impact—regardless of actual outcomes—may be a significant and underrecognized driver of democratic attitudes and engagement.

Results

Study 1 (N = 37,079) analyzed representative survey data from the European Union across 38 European countries in 2021 (12). Respondents rated whether they perceived that AI and automation will eliminate more jobs than they will create (1 = Totally disagree to 5 = Totally agree). The average response (M = 3.16, SD = 1.16) was significantly above the scale's neutral midpoint, indicating that the public tends to perceive AI as more labor-replacing than labor-creating [one-sample t test: $t(35{,}749) = 25.34$, $P < 0.001$, $d = 0.13$, 95% CI (0.12, 0.14)].

We examined the robustness and cross-national generalizability of this finding. In 25 of the 38 countries, approximately two-thirds, the average perceived labor impact of AI was significantly above the scale midpoint, with small-to-moderate effect sizes (Ps ≤ 0.039, Bonferroni-corrected; ds: 0.12 to 0.51; see *Left* panel of Fig. 1). For a more refined assessment, we grouped countries by GDP per capita and found the same pattern in the top three quartiles [$d_{Q4} = 0.18$, 95% CI (0.16, 0.20); $d_{Q3} = 0.20$, 95% CI (0.18, 0.22); $d_{Q2} = 0.20$, 95% CI (0.17, 0.22)], but not in the lowest [$d_{Q1} = –0.02$, 95% CI (–0.04, 0.00)]. To assess temporal stability, we analyzed a follow-up dataset again collected by the European Union three years later (N = 26,404; 27 countries), which replicated the original pattern (see *SI Appendix*).

Fig. 1.



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Public perceptions of AI as labor-replacing (vs. labor-creating) by country (*Left* panel) and by GDP per capita quartile (*Right* panel) in Study 1 (N = 37,079). Colored dots and whiskers indicate the observed standardized mean differences (Cohen's d) from the 1 to 5 scale midpoint and their 95% CI.

Next, we related these perceptions to our two democratic outcomes of interest: democratic legitimacy (measured using satisfaction with democracy in respondents' home country; 1 = Not at all satisfied to 4

= Very satisfied; Ref. 9) and political engagement with technology (measured using four variables; e.g., "Do you attend public meetings or debates about science and technology"; 1 = No, never to 4 = Yes, regularly; α = 0.85). We estimated separate multilevel regression models with both outcomes as dependent variables, public AI labor market perceptions as the independent variable, and technology-related, political, and sociodemographic controls at the individual and country level. Dependent variables were log-transformed, allowing coefficients to be interpreted as approximate percentage changes. Across all models, perceiving AI as labor-replacing (vs. labor-creating) was significantly associated with lower satisfaction with democracy (see Models 1A–3A; Table 1) and lower political engagement with technology (see Models 1B-3B; Table 1). The estimated associations were small-to-moderate (Ref. 13; see *SI Appendix*); for instance, a one-point increase in perceiving AI as labor-replacing (vs. labor-creating) was associated with approximately 5.0 to 2.5% lower satisfaction with democracy across all models—a relationship of similar order of magnitude to being unemployed (see Model 2A) and to established predictors such as personal income satisfaction (14). Multiple robustness checks supported these findings (*SI Appendix*). Models with country fixed effects, a composite index averaging respondents' satisfaction with democracy in their home country and in the European Union, and untransformed dependent variables produced consistent estimates (all *P*s < 0.001). Sensitivity analyses indicated that the findings were robust to unobserved confounding—for example, an omitted variable three times as strongly related to both AI perceptions and satisfaction with democracy as life satisfaction would still leave the relationship significant ($RV_{q=1,\ a=0.05}$ = 6.76%). Finally, the follow-up EU dataset collected three years later (N = 26,404) replicated the negative association between satisfaction with democracy and perceiving AI as labor-replacing—both in separate analyses (all *P*s < 0.01) and in pooled mixed-effects models accounting for time- and country-level variation (all *P*s < 0.001).

Table 1.

Multilevel regression models of satisfaction with democracy and political engagement with technology (Study 1)

	Dependent variable: Satisfaction with democracy[a]			Dependent variable: Political engagement with technology[a]		
Model	**1A**	**2A**	**3A[b]**	**1B**	**2B**	**3B[b]**
Perceptions of AI as labor-replacing (vs. labor-creating)	–0.050***	–0.049***	–0.025***	–0.054***	–0.054***	–0.038***
	(0.004)	(0.004)	(0.002)	(0.005)	(0.005)	(0.004)
Unemployed		–0.069***	–0.021		–0.017	–0.007
		(0.015)	(0.014)		(0.012)	(0.035)
Positive attitude toward technology			0.007**			0.019***
			(0.003)			(0.005)
Scientific literacy			0.035***			0.002
			(0.008)			(0.011)
Political orientation (left–right)			0.008			–0.004
			(0.005)			(0.003)
Right direction country			0.164***			0.008*
			(0.009)			(0.005)
Life satisfaction			0.096***			–0.007
			(0.009)			(0.005)
Constant	1.028***	1.030***	0.124	0.499***	0.500***	0.342
	(0.024)	(0.024)	(0.188)	(0.026)	(0.026)	(0.346)
Observations	35,544	35,544	26,981	35,262	35,262	26,816

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Notes: All models include country random effects. Coefficients represent unstandardized regression coefficients (robust SE in parentheses). [a]Log-transformed. [b]Models 3A and 3B additionally include individual-level controls (age, gender, education, occupation, and technology-related media consumption) and country-level controls (government effectiveness, GDP per capita, and egalitarianism); see *SI Appendix* for full results. ***$P < 0.01$, **$P < 0.05$, and *$P < 0.1$.

These results suggest that the public in developed countries views AI as labor-replacing and that these perceptions are negatively associated with key political outcomes. However, because these findings are correlational, they do not allow for causal conclusions. Study 2 addresses this limitation using an experimental design.

Study 2 employed a nationally representative UK sample ($N = 1{,}202$), one of the sample countries in Study 1. Participants were randomly assigned to one of two conditions (AI frame: labor-replacing vs. labor-creating; between-participants). In the AI labor-replacing frame condition, participants were asked to imagine that in the future AI will eliminate more jobs than it creates, mirroring the EU survey phrasing from Study 1. In the AI labor-creating frame condition, participants read the opposite. We measured our two dependent variables on 7-point Likert-scales (1 = Strongly disagree to 7 = Strongly agree): erosion of trust in democracy (three items; e.g., "In a world where AI eliminates (vs. creates) more jobs than it creates (vs. eliminates), I would lose trust in existing democratic institutions."; $\alpha = 0.98$), an important indicator of democratic legitimacy (15), and willingness to politically engage with future AI developments (three items; e.g., "When AI eliminates (vs. creates) more jobs than it creates (vs. eliminates), I would actively participate in shaping future developments on AI."; $\alpha = 0.89$). Additionally, we measured believability of the AI frame (1 = Not at all believable to 7 = Very much believable) and political orientation (1 = Left to 10 = Right) as exploratory moderators. We also included attention and manipulation checks (*SI Appendix*).

Consistent with the results of Study 1, participants exposed to the AI labor-replacing frame reported significantly greater erosion of trust in democracy than those exposed to the AI labor-creating frame [$M_{replace} = 4.60$, $SD_{replace} = 1.71$ vs. $M_{create} = 2.83$, $SD_{create} = 1.56$; $t(1{,}200) = 18.84$, $P < 0.001$, $d = 1.09$, 95% CI (0.97, 1.21)] and lower willingness to politically engage with future AI developments [$M_{replace} = 3.55$, $SD_{replace} = 1.64$ vs. $M_{create} = 4.44$, $SD_{create} = 1.64$; $t(1{,}200) = -9.36$, $P < 0.001$, $d = -0.54$, 95% CI (–0.66, –0.42)]. Both effects became stronger when participants perceived the AI frame as more believable (moderation effect: $b_{erosion\ trust} = 0.218$, $P < 0.01$; $b_{political\ engagement} = -0.261$, $P < 0.001$), yet remained robust: the AI labor-replacing frame increased erosion of trust in democracy across all believability levels (all Ps < 0.001), and reduced willingness to politically engage for scores above 1

(all $Ps < 0.01$). Political orientation did not moderate either effect (all $Ps \geq 0.154$). A replication with a representative US sample ($N = 1{,}200$) produced consistent results (*SI Appendix*).

These findings provide causal evidence that framing AI as labor-replacing (vs. labor-creating) increases erosion of trust in democracy and reduces political engagement.

Discussion

Experts debate AI's economic implications, yet little research has examined how the public perceives this debate—and how such perceptions shape democratic attitudes. Prior work has focused on how people directly exposed to automation change their political behavior (e.g., vote choice; Ref. 3). Using complementary cross-national correlational and experimental evidence, this work shows that the public tends to view AI as labor-replacing and that this perception—independent of actual outcomes—can decrease democratic legitimacy and political engagement in shaping AI's future. These effects may be particularly concerning at a time when democratic legitimacy is declining across many established democracies (9) and democratic input in AI's technological trajectory and labor impact is critical (11).

Our effects were robust across European and US contexts and levels of the preregistered moderators perceived believability and political orientation (see Study 2). Future research should examine additional moderators, measure them pretreatment to avoid biasing moderation analyses (unlike Study 2), and test potential mediators to clarify psychological processes. For instance, perceiving AI as labor-replacing may elicit symbolic status threats or reduce perceived political efficacy, undermining democratic attitudes and engagement without realized economic harm. Study 2, which manipulated the AI frame, produced sizable effects (erosion of trust: $d = 1.09$; political engagement: $d = -0.54$). This suggests that public beliefs of AI's labor impact—and their downstream consequences—are malleable (e.g., through targeted communication), an important avenue for future research. Finally, while our satisfaction-with-democracy measure captures an important indicator of democratic legitimacy, it does not necessarily reflect commitment to democratic principles (9). Future work should link public AI labor market perceptions to more diffuse forms of democratic support (e.g., preferences for undemocratic alternatives).

Our findings underscore the need for policymakers and researchers to look beyond AI's technical progress. Public narratives about AI—and particularly its perceived threat to employment—may generate political consequences before economic ones materialize.

Materials and Methods

Methodological details and full variable wording are provided in the *SI Appendix*.

Study 1 (N = 37,079; M_{age}= 48.05, SD_{age} = 17.27; 52% female) draws on a representative survey from 38 European countries (April–May 2021; Ref. 12).

Study 2 draws on a nationally representative UK sample (N = 1,202; M_{age}= 46.97, SD_{age} = 15.73; 52% female) obtained via Prolific (https://www.prolific.com/). It was preregistered (https://aspredicted.org/96k4ha.pdf) and approved by the LMU School of Management Ethics Committee. Informed consent was obtained from all participants.

Supplementary Material

Appendix 01 (PDF)

pnas.2523508123.sapp.pdf (315.6KB, pdf)

Acknowledgments

Author contributions

A.G., A.R., and C.F. designed research; A.G., A.R., and C.F. performed research; A.G. and A.R. analyzed data; and A.G., A.R., and C.F. wrote the paper.

Competing interests

The authors declare no competing interest.

Data, Materials, and Software Availability

Datasets, analysis code, preregistrations, and study materials have been deposited in the Open Science Framework (https://doi.org/10.17605/OSF.IO/XZU3V) (16).

Supporting Information

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

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pnas.2523508123.sapp.pdf (315.6KB, pdf)

Data Availability Statement

Datasets, analysis code, preregistrations, and study materials have been deposited in the Open Science Framework (https://doi.org/10.17605/OSF.IO/XZU3V) (16).

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